Assigment 3

Question 1

a

To check cheat and pass conditinally independent given study,

we need to,

$$P(\text{cheat} \cap \text{Pass} \mid \text{study}) = P(\text{cheat} \mid \text{study}) * P(\text{Pass} \mid \text{study})$$

L.H.S

$$P(\text{cheat} \cap \text{Pass} \mid \text{study}) = \frac{P(\text{cheat} \cap \text{Pass} \cap \text{study})}{P(\text{study})}$$

$$= \frac{.25}{.25 + .15 + .02 + .03}$$

$$= 0.56$$

R.H.S

$$P(\text{cheat} \mid \text{study}) = \frac{P(\text{cheat} \cap \text{study})}{P(\text{study})}$$

$$= \frac{.25 + 0.02}{0.45} = 0.6$$

$$P(\text{Pass} | \text{Study}) = \frac{P(\text{Pass} \cap \text{Study})}{P(\text{Study})}$$

$$= \frac{0.25 + 0.15}{0.45} = 0.89$$

$\therefore P(\text{Cheat} | \text{Study}) * P(\text{Pass} | \text{Study}) = 0.6 \times 0.89 = 0.534$

$\therefore$ L.H.S $\neq$ R.H.S

They are not conditionally independent given study.

b

P(Pass or cheat)

$$P(\text{Pass} \cup \text{cheat}) = P(\text{Pass}) + P(\text{cheat}) - P(\text{Pass} \cap \text{cheat})$$

$$P(\text{Pass}) = 0.25 + 0.15 + 0.10 + 0.13 = 0.63$$

$$P(\text{cheat}) = 0.25 + 0.02 + 0.10 + 0.22 = 0.59$$

$$P(\text{Pass} \cap \text{cheat}) = .25 + .10 = .35$$

$$P(\text{Pass} \cup \text{cheat}) = 0.63 + 0.59 - 0.35$$

$$= 0.87$$

Question 2

a

The marginal probability of ¬cold

$P(\neg cold) = .12 + .04 + .10 + .07$

$= 0.33$

b

Not cloudy given not raining and weather being not cold

$P(\neg cloudy \mid \neg Rain \cap \neg cold)$

$= \dfrac{P(\neg cloudy \cap \neg Rain \cap \neg cold)}{P(\neg Rain \cap \neg cold)}$

$= \dfrac{0.7}{0.04 + 0.07} = 0.636$

$= 0.64$

c

not rain given not cloudy

$P(\neg Rain \mid \neg Cloudy)$

$= \dfrac{\cancel{P(Rain}\ P(\neg Rain \cap \neg Cloudy)}{P(\neg Cloudy)}$

$= \dfrac{0.03+0.07}{.26+.03+.10+.07}$

$= \dfrac{0.1}{0.46} = 0.217 \approx 0.22$

d

$P(\neg Rain \text{ or } cloudy)$

$P(\neg Rain \cup cloudy) = P(\neg Rain) + P(cloudy) - P(\neg Rain \cap cloudy)$

$P(\neg Rain) = \cancel{0.6+0.4+0.3+}\ .06+.04+.03+.07$

$= 0.2$

$P(cloudy) = .32+.12+.06+.04 = 0.54$

$P(\neg Rain \cap cloudy) = 0.06 + 0.04 = .10$

$\therefore P(\neg Rain \cup cloudy) = 0.2 + 0.54 + .10 = 0.64$

Question 3

a

$$P(Football | Left\text{-}Hand) = \frac{P(football \cap Left\text{-}Hand)}{P(Left\text{-}Hand)}$$

$$= \frac{0.15}{0.24 + 0.15 + 0.15} = 0.2778$$

b

$$P(Right\text{-}Handed | Cricket) = \frac{P(cricket \cap Right\ Hand)}{P(cricket)}$$

$$= \frac{.10}{0.24 + 0.10}$$

$$= 0.29$$

c

$P(\text{football} \cap \text{cricket}) = 0$

Because they are mutually exclusive.

d

$P(\text{Right Hand} \cup \text{Left Hand}) = P(\text{Right hand}) + P(\text{Left hand})$
$\qquad - P(\text{Right hand} \cap \text{Left hand})$

$= P(0.1 + 0.1 + .26) + P(.24 + .15 + .15)$
$- 0$

$= 1$

e

$P(\text{Football} \cap \text{Right Hand}) = 0.10$

$P(\text{Football}) \times P(\text{Right Hand}) = (0.15 + .10) \times 0.46 = 0.115$

$P(\text{football} \cap \text{Right Hand}) \neq P(\text{Football}) \times P(\text{Right Hand})$

so they are dependent.